

The Stem

Origin and Structure

The stem is the ascending axis, developing from the **plumule**. It bears branches, leaves, flowers and fruits.

Nodes and Internodes

Stems are distinguished from roots by **nodes** (points bearing leaves) and **internodes** (segments between two nodes).

Buds

Stems bear **terminal (apical)** buds and **axillary** buds. Young stems are green and herbaceous; mature stems often become woody.

Functions

Spreading branches for light interception, conduction of water/minerals/photosynthates, and modification for storage, support, protection or propagation.

Underground Stem Modifications

Besides tubers, stems modify into **rhizomes** (ginger, turmeric) that grow horizontally underground storing food and enabling propagation, and **corms** (Colocasia) which are short, vertical, swollen stems.

Aerial Stem Modifications

Aerial modifications include **stem tendrils** (grapevine, for climbing), **thorns** (Citrus, Bougainvillea, for protection), and **phyllodes** (Opuntia) where flattened green stems perform photosynthesis in place of leaves.

Sub-aerial Stem Modifications

Sub-aerial stems creep along the ground and help in vegetative propagation, including **runners** (grass), **stolons** (mint), **offsets** (Pistia), and **suckers** (banana, chrysanthemum).

Stem vs Root — Key Differences

Feature	Stem	Root
Origin	Plumule	Radicle
Nodes/Internodes	Present	Absent
Geotropism	Negatively geotropic	Positively geotropic
Chlorophyll	Often present	Usually absent

Functions of the Stem

- Support & branching
- Conduction of water/minerals/food
- Food storage (modified stems)
- Vegetative propagation